

HR Employee Attrition & Workforce Analysis

1. Project Overview

This project analyzes workforce and attrition patterns for a 10,000-employee organization using HR records covering demographics, compensation, performance, and engagement metrics. The goal is to identify the key drivers behind employee attrition, understand pay and satisfaction trends across departments and job levels, and provide data-driven recommendations to help HR improve retention.

2. Dataset Summary

- Rows: 10,000 - Columns: 15 : - Key Features:

- Employee profile (Employee ID, Age, Department, Job Level, Age Group)
 - Tenure & compensation (Years at Company, Monthly Income, Promotion in Last 5 Years)
 - Engagement & performance (Job Satisfaction, Work-Life Balance, Performance Rating, Training Hours)
 - Work conditions (Overtime, Distance From Home) and outcome (Attrition)
- Missing Data: 0 null values across all columns - Duplicates: none found

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- Data Loading:** Imported the dataset using `pandas`.
- Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

	employeeid	age	department	joblevel	yearsatcompany	monthlyincome	jobsatisfaction	worklifebalance
count	10000	10000.000000	10000	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000
unique	10000	NaN	7	NaN	NaN	NaN	NaN	NaN
top	EMP0000	NaN	Sales	NaN	NaN	NaN	NaN	NaN
freq	1	NaN	2453	NaN	NaN	NaN	NaN	NaN
mean	NaN	37.693500	NaN	2.380800	7.788400	8967.918800	3.362100	2.761500
std	NaN	9.376457	NaN	1.141368	7.141941	3500.745748	1.059951	0.893027
min	NaN	22.000000	NaN	1.000000	0.000000	3000.000000	1.000000	1.000000
25%	NaN	31.000000	NaN	1.000000	2.000000	6043.500000	3.000000	2.000000
50%	NaN	37.000000	NaN	2.000000	6.000000	8670.000000	3.000000	3.000000
75%	NaN	44.000000	NaN	3.000000	12.000000	11484.000000	4.000000	3.000000
max	NaN	60.000000	NaN	5.000000	35.000000	21410.000000	5.000000	4.000000

`df.head()` — first 6 rows of the cleaned dataset

	age	joblevel	yearsatcompany	monthlyincome
count	10000.0	10000.0	10000.0	10000.0
mean	37.69	2.38	7.79	8967.92
std	9.38	1.14	7.14	3500.75
min	22.0	1.0	0.0	3000.0
25%	31.0	1.0	2.0	6043.5
50%	37.0	2.0	6.0	8670.0
75%	44.0	3.0	12.0	11484.0
max	60.0	5.0	35.0	21410.0

	jobsatisfaction	worklifebalance	distancefromhome	performancerating	traininghourslastyear
count	10000.0	10000.0	10000.0	10000.0	10000.0
mean	3.36	2.76	9.91	2.6	14.99
std	1.06	0.89	6.97	0.74	3.89
min	1.0	1.0	1.0	1.0	2.0
25%	3.0	2.0	4.7	2.0	12.0
50%	3.0	3.0	8.3	3.0	15.0
75%	4.0	3.0	13.4	3.0	18.0
max	5.0	4.0	50.0	4.0	31.0

df.describe() — summary statistics for numeric columns

- **Missing Data Handling:** Confirmed with `df.isnull().sum()` that all 15 columns are fully populated — no imputation required.
- **Column Standardization:** Renamed columns to **snake_case** for consistency and easier scripting.

Feature Engineering:

- Created `age_group` column by splitting employee ages into three equal-sized bins (Adult, Middle age, Senior) using `pd.qcut()`.
- Verified `df.duplicated().sum()` returned 0 — no duplicate employee records.

- **Data Types Check:** Used `df.dtypes` and `df.nunique()` to confirm categorical vs. numeric fields before modeling in Power BI.

4. Workforce & Attrition Analysis using Python (Pandas)

We grouped and aggregated the cleaned data in pandas to answer key HR business questions:

1. Attrition by Department – Compared headcount and attrition rate across all seven departments.

department	total_employees	attrition_count	attrition_rate_%
Engineering	2045	196	9.58
Sales	2453	233	9.5
Marketing	1188	108	9.09
IT	1035	93	8.99
Operations	1513	135	8.92
Finance	1037	85	8.2
HR	729	53	7.27

2. Attrition by Job Level – Checked whether attrition is more common among junior or senior employees.

job_level	total_employees	attrition_count	attrition_rate_%
1.0	2844.0	319.0	11.22
2.0	2627.0	241.0	9.17
3.0	2773.0	254.0	9.16
4.0	1389.0	70.0	5.04
5.0	367.0	19.0	5.18

3. Average Monthly Income by Department – Compared average pay across departments to spot compensation gaps.

department	avg_monthly_income
Engineering	10514
IT	10183
Finance	9723
Marketing	8902
Sales	8255
Operations	7654
HR	7069

4. Overtime vs. Attrition – Measured how working overtime relates to the likelihood of leaving.

overtime	stayed	left	attrition_rate_%
No	6595	556	7.78
Yes	2502	347	12.18

5. Job Satisfaction Distribution – Reviewed how employees rated their job satisfaction (1 = lowest, 5 = highest).

job_satisfaction_score	employee_count
1	500
2	1455
3	3476
4	3062
5	1507

6. Promotion (Last 5 Years) vs. Attrition – Checked whether employees who were promoted recently are less likely to leave.

promoted_last_5_years	stayed	left
No	8244	842
Yes	853	61

7. Work-Life Balance Distribution – Reviewed how employees rated their work-life balance (1 = lowest, 4 = highest).

work_life_balance_score	employee_count
1	983
2	2515
3	4406
4	2096

5. Dashboard in Power BI

Finally, we built an interactive HR Analytics dashboard in Power BI to present these insights visually. The report page includes five headline KPI cards, department / job level / overtime slicers, and six cross-filterable visuals covering attrition, compensation, and engagement.

HR ANALYTICS DASHBOARD



Power BI — HR Analytics Dashboard

6. Business Recommendations

- **Target Overtime Reduction** – Employees working overtime leave at a noticeably higher rate than those who don't; review workload distribution in high-overtime teams.
- **Support Junior Staff** – Job Level 1 employees have the highest attrition rate of any level; introduce structured onboarding and early-career mentorship.
- **Invest in Promotion Pathways** – Employees who were promoted in the last 5 years stay at a noticeably higher rate; build clearer, faster promotion tracks.
- **Review Departmental Pay Gaps** – Average income varies meaningfully across departments; benchmark Sales, Operations and HR pay against market rates.
- **Strengthen Work-Life Balance** – A meaningful share of employees rate work-life balance at 1 or 2; expand flexible-work and wellness initiatives.
- **Prioritize High-Risk Departments** – Engineering and Sales show the highest attrition rates; focus retention interviews and stay-bonuses there first.